Descriptive Logic IV: The Rough Ground

Philosophical Investigations §§89–133, in Phient's terms

Phiano Project · `bin/family --rough` · the *Phi-4's Families* page · 2026-09-30

> In Phient's world, a word (w) is nothing but what it relates to, under what conditions (c). The Phical paradigm, X ⇒ Y, drives and composes Phient's Phasor Language Models: Phinet, Phiano, PhiTon. Phical stores no point: a word is 32–64 relations, each a complex phasor on a closed circle with a count; meaning is a side effect of pairs of words moving under a condition.

§§89–133 turn from language to philosophy itself. Philosophy is to describe, not explain. It is a fight against being bewitched by our own forms of expression. It needs friction: back to the rough ground. It brings things into a view that can be surveyed at once, leaves everything as it is, and has methods rather than one method. In our project the philosophy is the code, so this chapter turns the remarks on the work of Chapters I–III. It asks three questions:

1. Does the teacher's language meet the ideal of a consistent calculus?
2. Can its use be put in plain view?
3. Is there one method that wins?

As before, Wittgenstein is paraphrased and cited by remark, never quoted.

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1. Three more definitions

- **D14 · Rough ground.** The teacher's answers are the ground. Read by the kinds' ideal meaning, some of them contradict each other. A descriptive learner takes them as use and does not repair them first (§107, §125).
- **D15 · Surveyable view.** A representation is surveyable when it lets the connections be seen at once, including the *intermediate cases*. For Phient, that means the posts table (Chapter I), the fibre chart and the words that bridge families (the *Phi-4's Families* page) (§122).
- **D16 · Methods, not a method.** No single learner is best on every test. Each works where its form of reading matches the game (§133).

2. The remarks, in Phient's terms

§§89–92 — The sublime, and what lies hidden

The remarks. Logic seems to have a peculiar depth, as if it reached to the essence of everything (§89). We feel we must see *into* phenomena, when our inquiry is about the kinds of statement we make about them (§90). This can give the idea of a final, completely exact analysis still waiting to be found (§91). Then the essence seems hidden, something that lies beneath the surface (§92).

In Phient's terms. Nothing in Chapters I–III lies beneath the surface. A post is a count anyone can read. A family is overlaps anyone can draw. A word's identity is spread over relations anyone can list. The functions are the one place something is not on the surface. Chapter III found they carry what the counts cannot (descriptions never told). But what they carry is still *what the word relates to*, read back through the operator. There is no second, deeper meaning in them.

§§93–100 — The ideal must be in reality

The remarks. A proposition seems a remarkable thing. We are tempted to sublime the logic of our language (§§93–94). Thought seems unique, the order of the world's possibilities laid down in it (§§95–97). Yet every sentence of our language is in order as it is. We are not striving after an ideal, as if ordinary vague sentences had no sense yet (§98). A vague sense seems to be no sense at all, like an enclosure with a gap. But an enclosure with a hole in it is still an enclosure (§§99–100).

In Phient's terms. Chapter III measured the enclosure with a hole. The post threshold works across a range (0.5–1 nat). A post is a boundary with a blur, and it serves. Every episode of the builders' game was in order as it was; the learners did not need a cleaner language first.

§§101–108 — Crystalline purity, and friction

The remarks. We think the ideal must be found in reality, since we think we already see it there (§101). The strict, clear rules of logic seem to lie in the background, hidden in the medium of understanding (§102). The ideal is fixed in our thinking like glasses on our nose, through which we see whatever we look at (§103). The crystalline purity of logic was not a *result* of investigation; it was a *requirement* (§107). The conflict became unbearable. We have got onto slippery ice where there is no friction, so in a sense the conditions are ideal — but we cannot walk. We want to walk, so we need friction: back to the rough ground (§107). The preconception of crystalline purity can only be removed by turning the whole inquiry around (§108).

In Phient's terms (D14): measured. Read by the kinds' ordinary meaning, some of Phi-4's answers contradict each other. `bin/family --rough` counts them in the night table:

contradiction	rows	examples
a is-a b, and b is-a a	39	animal is-a mammal (and mammal is-a animal); car / automobile
a has b, and a part-of b	164	appendage has limb, and appendage part-of limb
a part-of b, and b part-of a	12	computer part-of device, and device part-of computer
total	**210 of 3,630 rows (5.8%)**	

Some of these are errors, such as animal is-a mammal. Many are the teacher using *is-a* for "is another word for" (car and automobile), or *has* and *part-of* loosely. So the "contradiction" belongs to our ideal reading of the kinds, not to the teacher's use.

Does it stop the game? phinet is trained on the told rows with the contradicting ones and without them. Both models are read on the same 702 held-out rows, none of which contradicts anything. Four seeds:

seed	with: MRR / within kind	without: MRR / within kind
7	0.084 / 0.111	0.080 / 0.112
8	0.075 / 0.107	0.076 / 0.107
9	0.073 / 0.128	0.076 / 0.123
10	0.083 / 0.114	0.094 / 0.127
mean	**0.079 ± 0.006 / 0.115**	**0.082 ± 0.009 / 0.117**

The difference is smaller than the spread between seeds. Removing the contradictions does not measurably help. This fits docs/101: *imposing* declared laws (the converse, transitivity) cost 0.016–0.027 MRR. Purity as a requirement costs something. Leaving the ground rough costs nothing we can measure.

§§109–115 — Description, and a picture that held us captive

The remarks. Our considerations must not be scientific ones. We may not advance any kind of theory. All *explanation* must disappear, and description alone must take its place. Philosophy is a battle against the bewitchment of our intelligence by means of language (§109). A picture held us captive, and we could not get outside it, because it lay in our language, which seemed to repeat it to us relentlessly (§115).

In Phient's terms. Our trial logs record where a picture held the *code* captive:

- **Trial 1**, summed composition, was the picture that every word signifies something under every aspect.
- **Trial 8**, agreement judged over the whole language, was the picture of one global rule where there are only local games.
- **Trial 3**, which measured a function against no preference, was the picture that a function's reading is a probability like a count's.

Each was fixed by describing more closely what the episodes showed. We did not add a theory.

§§116–121 — Back to everyday use

The remarks. When philosophers use a word such as *knowledge*, *being* or *object* and try to grasp its essence, we must ask whether the word is ever used that way in the language-game that is its home. We bring words back from their metaphysical to their everyday use (§116). The results of philosophy are the bumps the understanding gets by running its head up against the limits of language (§119). When I talk about language, I have to speak everyday language (§120).

In Phient's terms. A word's home is the game where it holds its posts. D3 carries a word only into games that agree with its home on an aspect. This is §116 as an operation: a word read outside its home, where the games disagree, falls silent, and the condition answers.

§§122–124 — The surveyable view; leaving everything as it is

The remarks. A main source of our failure to understand is that we do not *command a clear view* of the use of our words. A surveyable representation produces the understanding that consists in seeing connections, hence the importance of finding and inventing *intermediate cases* (§122). A philosophical problem has the form: I don't know my way about (§123). Philosophy may in no way interfere with the actual use of language. In the end it can only describe it, and it leaves everything as it is (§124).

In Phient's terms (D15). The *Phi-4's Families* page is a surveyable view of the teacher's categories. For each of the 12 families it shows:

- the fibre chart: which member holds which relation, with no row full;
- who shares with whom;
- the words that bridge families.

The bridges are the intermediate cases: *engine* is filed under component, device and machine; *dog*, *feline* and *monkey* under both animal and mammal. The posts table of Chapter I is the same kind of view for the builders' language.

§124 is the stance of the whole series. The learners describe Phi-4's use and do not correct it. The one time Phient interfered, with declared laws imposed as structure (docs/101), the student fitted the teacher's answers worse.

§§125–128 — The contradiction, and what everyone would admit

The remarks. Philosophy's business is not to resolve a contradiction by a mathematical discovery. It is to get a clear view of the state of affairs *before* the contradiction is resolved. We lay down rules for a game, then find that when we follow them things do not turn out as we assumed. We become entangled in our

own rules (§125). Philosophy simply puts everything before us (§126). Its work is assembling reminders for a particular purpose (§127). If one tried to advance *theses* in philosophy, everyone would agree to them (§128).

In Phient's terms. The 210 contradicting rows are laid out in plain view (docs/106), not resolved. They cost nothing measurable. Our own entanglement in our own rules is the trial log: the posts gate, the game-agreement rule and the condition prior were each a rule we laid down that did not turn out as assumed.

§§129–133 — Objects of comparison; methods, like therapies

The remarks. The aspects of things that matter most to us are hidden by their simplicity and familiarity (§129). Our clear, simple language-games are not preliminary studies toward a future regularisation of language. They are *objects of comparison*, meant to throw light on our language by similarities and dissimilarities (§130). We avoid dogmatism only by presenting the model as what it is, an object of comparison, a measuring rod (§131). We want to establish an order for a particular purpose, one of many possible orders (§132). There is not *a* philosophical method, though there are methods, like different therapies (§133).

In Phient's terms. The builders' game and the squares game are exactly this: objects of comparison for Phi-4's language, not models of it. Chapter III measured Phi-4's own answers against what the games taught us to look for (families, cluster names). The next measuring rod is Phi-4 playing the builders' game itself.

D16, as a table: which method does what best, collected from Chapters I–IV:

what is asked	best	not good at it	source
act from a known language, enough data	bag of words = counts = phinet (100%)	naming table (64%), lookup	ch. I
learn from fewest episodes	bag of words (95–100% at 160)	phinet (55–65% at 160)	ch. I
a word met once, pointed at, with the word `colour`	counts, phinet (100%)	bag of words (0%)	ch. I
composite words in new places	Phient reader (100%)	positional softmax (≤ 5%)	ch. II
a teacher who errs, errors marked	counts (100%)	bag of words (89–95%)	ch. II
a word from descriptions never told	phinet functions (25–28%)	counts (0%)	ch. III
unseen relations of Phi-4's words	phinet + inheritance (0.112 held-out MRR vs prior 0.053)	counts (no answer)	docs/99
a contradicting teacher	all unchanged	—	ch. IV

No row has the same winner as every other row. This is D16, and it is also the design of Phient: counts first, functions for what the counts cannot reach, and a condition that stays local to its game.

3. What Chapter IV adds

- **The teacher's language is rough ground, and that is fine.** 5.8% of Phi-4's answers contradict each other under the ideal reading of the kinds. Removing them does not measurably help (0.079 → 0.082 MRR, inside the seed spread). Imposing an ideal as structure did cost (docs/101).
- **A surveyable view of the teacher's categories**, as a page, with the intermediate cases in plain view.
- **Methods, not a method,** gathered into one table from four chapters.

4. Next

This ends the part of the *Investigations* that runs to §133. The next part, §§134–242 and above all §§185–242, is about following a rule. It asks what makes an answer the *right* continuation, when any finite set of examples fits many rules. Phient's version of that question is its central one: when the student continues where the teacher's examples stop, what makes the continuation right? That chapter will need Phi-4 itself as the other player, so the run on the GPU machine comes first (`scripts/phi4_builders.ps1`).

Code: `phiano/src/bin/family/rough.rs` (`--rough`). *Page: Phi-4's Families* (a published artifact). *Raw tables:* docs/106.

Run: `cargo run --release --bin family -- data/quest/night_2026-09-29/all.tsv out.md --rough` (about 20 minutes, 4 seeds).